



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

School of Computer Science and Statistics

Week 4 Assignment

CS7CS2 – Optimisation for Machine Learning

Lecturer: Dr. Mouli Chakraborty

Contact: mchakrab@tcd.ie

General Guidelines

- You may discuss ideas with classmates; however, **all written work and code must be entirely your own**. Do not share code. All submissions will be checked for plagiarism.
- Reports must be typed and submitted as a **single PDF along with a single .ipynb or .py file via Blackboard**. Do not submit the report inside a zip file.
- **Important:** Marks are awarded primarily for explanation, reasoning, and interpretation, not for raw code or numerical output.
 - Code-based results must be explained.
 - Numerical results must be interpreted.
 - Statements such as “see code” or unexplained plots/numbers are insufficient.
- All plots must have labelled axes, readable text, and a clear written explanation of what they show.

Submission Rules

Submit:

- A short PDF report with codes, plots and brief explanations.
 - A .ipynb or .py file containing the executable Python code.
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Q1. Consider the function

$$f(x, y) = x^2 + 100y^2$$

with initial condition $(x_0, y_0) = (2, 2)$. Run 200 iterations.

(I) Apply the following algorithms:

- Polyak step size
- RMSProp ($\alpha = 0.2, \beta = 0.9$)
- Heavy Ball ($\alpha = 0.01, \beta = 0.9$)
- Adam ($\alpha = 0.1, \beta_1 = 0.9, \beta_2 = 0.999$)

Choose reasonable baseline parameters. Plot the function value vs iteration for all four methods on one graph.

(II) For Heavy Ball :

- Fix $\beta = 0.9$.
- Gradually increase α until instability occurs $\alpha = 0.006, 0.01, 0.02$.

Plot the x -coordinate vs iteration for the given set of α .

(III) Plot contour lines of $f(x, y)$ and overlay trajectories for Heavy Ball ($\beta = 0.9, \alpha = 0.01$), RMSProp ($\beta = 0.9, \alpha = 0.2$), and Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.1$).

Briefly explain why the function causes zig-zag behaviour, how momentum affects oscillations, how adaptive methods respond to different curvature in x and y , and why ill-conditioning influences the stability limit for α .

Q2. Consider the Rosenbrock function

$$f(x, y) = (1 - x)^2 + 100(y - x^2)^2$$

with initial condition $(x_0, y_0) = (-1.25, 0.5)$. Run 3000 iterations.

(I) Apply the following algorithms:

- Polyak step size ($\alpha \leq 0.1$)
- RMSProp ($\alpha = 0.01, \beta = 0.9$)
- Heavy Ball ($\alpha = 2e - 4, \beta = 0.9$)
- Adam ($\alpha = 0.05, \beta_1 = 0.9, \beta_2 = 0.999$)

Choose reasonable baseline parameters. Plot the function value vs iteration for all four methods on one graph.

(II) For Adam :

- Fix $\beta_1 = 0.9, \beta_2 = 0.999$.
- Gradually increase α until instability occurs $\alpha = 0.02, 0.05, 0.12$.

Plot the x -coordinate vs iteration for the given set of α .

(III) Plot contour lines of the Rosenbrock function and overlay trajectories for Heavy Ball ($\beta = 0.9, \alpha = 2e - 4$), RMSProp ($\beta = 0.9, \alpha = 0.01$), and Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \alpha = 0.05$).

Briefly explain why Rosenbrock has slow convergence for basic methods (narrow curved valley), how momentum changes oscillations/overshoot, how adaptive scaling affects progress along the valley, and why the stability limit for α is sensitive to curvature.